

COLLISION SERIES: 1M → 10M SCALING REPORT

A 70-Phase Causal Language Modeling Study in Extreme Low-Resource Regimes

Project: COLLISION	Target Scale: 1.46M to 10.28M Params
Hardware Regime: Consumer CPU (From Scratch)	Release Version: COLLISION-10M v1.0.0
Hugging Face Hub: collision-10M/collision-10m	Status: DATA_COLLECTION_HOLD (Phase 70)

1. Executive Summary

COLLISION is an open research investigation into Transformer causal language modeling operating under extreme resource constraints. The project demonstrates that high-quality data engineering, deterministic split protocols, and rigorous verification gates can produce stable, convergent representation spaces at sub-50M parameter scales on consumer CPUs without GPU pretraining.

Core Research Hypothesis: Data quality engineering, leakage elimination, and clean evaluation protocols drive model convergence more effectively than parameter expansion alone in low-resource regimes.

2. Evolution & Benchmark Progress (Phases 1 – 70)

Phase / Variant	Params	Val Perplexity	Key Technical Milestone
Phase 5 (Baseline)	1.46M	62.86	Severe overfitting & alphabetical split bias
Phase 6 (Data Audit)	1.46M	6.93	Dataset v4: Subject-wise split (0% leakage) dropped perplexity by 55.9 pts
Phase 10 (Scaling)	3.38M	3.75	Capacity expansion validation on CPU
Phase 13 (Instruct Test)	3.38M	Degraded	Synthetic instruction crash (Repetition 57.6%)
Phase 15 (Flagship Base)	10.28M	2.11	COLLISION-10M pretraining on 10M tokens (Test Perplexity: 1.79)
Phase 43 (DPO Audit)	10.28M	Loss: 3.19	Discovered missing reference ratio in DPO math
Phase 52 (SFT Candidate)	10.28M	Gen Score: 66.85	Apex Candidate J52 promoted (Coherence: 38.50)
Phase 70 (Final Release)	10.28M	0 Defects	Full telemetry & DATA_COLLECTION_HOLD verdict

3. Data Quality vs. Parameter Capacity Breakthrough

During Phase 5 pretraining of COLLISION-1.46M, validation perplexity stalled at 62.86. A comprehensive forensic audit revealed two structural errors: 1) Alphabetical document splitting caused complete domain missingness in training splits, and 2) 26.82% contiguous sentence leakage existed between train and val. In Phase 6, constructing collision_dataset_v4 with subject-wise deterministic splits reduced validation perplexity to **6.93** without changing model architecture.

4. The DPO Failure & Mathematical Bug Discovery

In Phase 42–46, Direct Preference Optimization (DPO) was explored. Initial DPO candidates suffered coherence collapse. Phase 43 forensic auditing discovered a critical code defect: the loss function omitted the reference model ratio π_{ref} , penalizing policy log-probs without KL anchoring. Even after repairing canonical DPO, experiments proved that standard DPO preference rewards penalize 10M models due to length/repetition sensitivity.

5. Supervised Fine-Tuning & Apex Candidate J52

Pivoting to Supervised Fine-Tuning (SFT) yielded the highest performance. Apex Candidate **J52** (Phase 52) trained on `collision_sft_v3` achieved: **Generalization Score: 66.85, Coherence: 38.50, Instruction Following: 48.20**.

6. Public Beta, Telemetry & Hugging Face Release

Phases 53–70 introduced a production FastAPI server, Streamlit client (COLLISION LAB), PII redaction engine, and automated readiness gates requiring 100 clean real-world human records. Per strict scientific guidelines, zero synthetic user data was generated, placing the project into an explicit `DATA_COLLECTION_HOLD_HUMAN_TRAFFIC_REQUIRED` status. COLLISION-10M v1.0.0 is officially published on Hugging Face at **collision-10M/collision-10m** with verified SHA256 `d256d46d...3775b97`.

Quick Hugging Face / Python Inference Example:

```
python release_inference.py --prompt "Artificial intelligence is" --checkpoint  
models/collision-10m/model.pt
```